Lecture 13 – CARTs and Ensemble Methods

[ISLP 8.1–8.2]

MC886 – Machine Learning

Marcelo S. Reis †

† Departamento de Sistemas de Informação
Instituto de Computação, Universidade Estatual de Campinas

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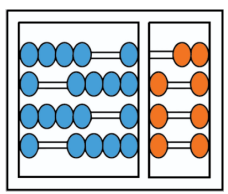

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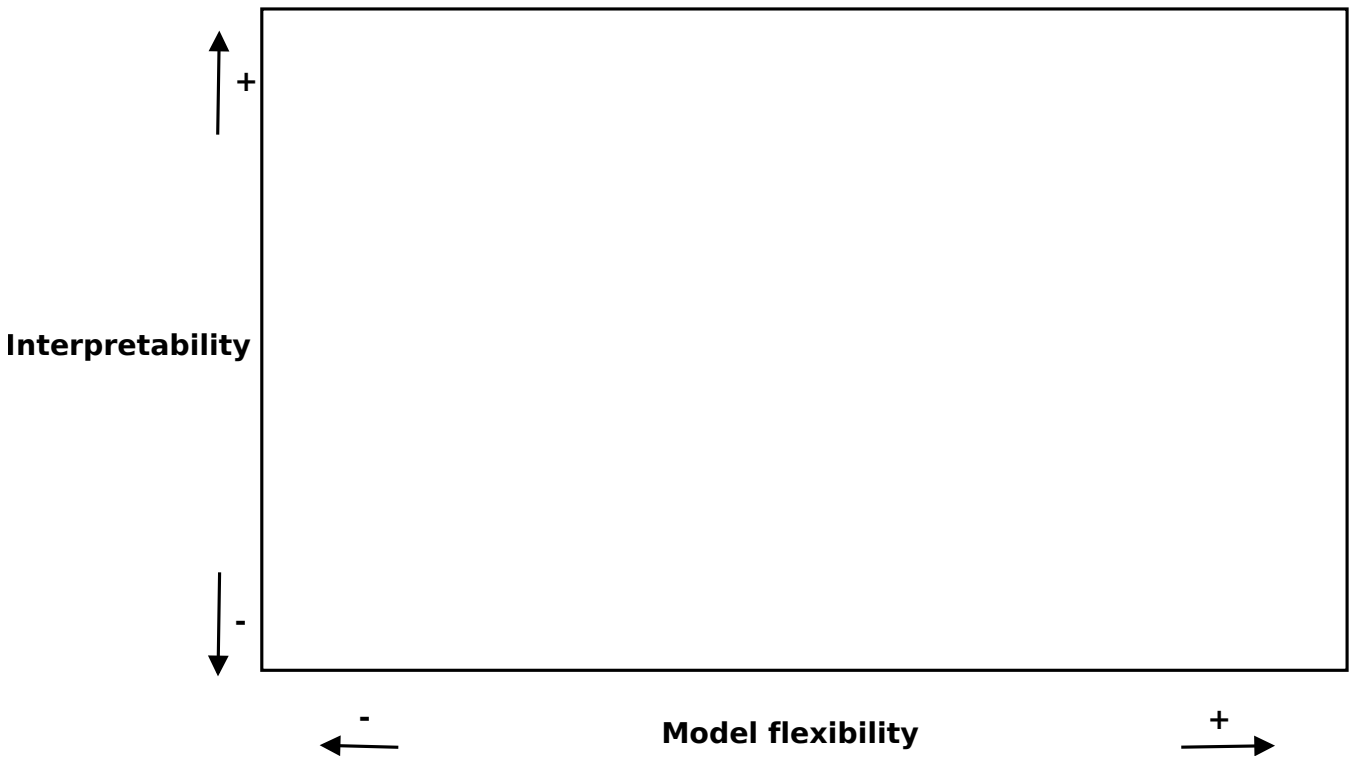
Today's summary

Classification and regression trees (CARTs)

Learning of CARTs

Ensemble learning: "Unity makes strengh!"

Bagging, Random Forest, Boosting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Interpretability
-
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Model flexibility
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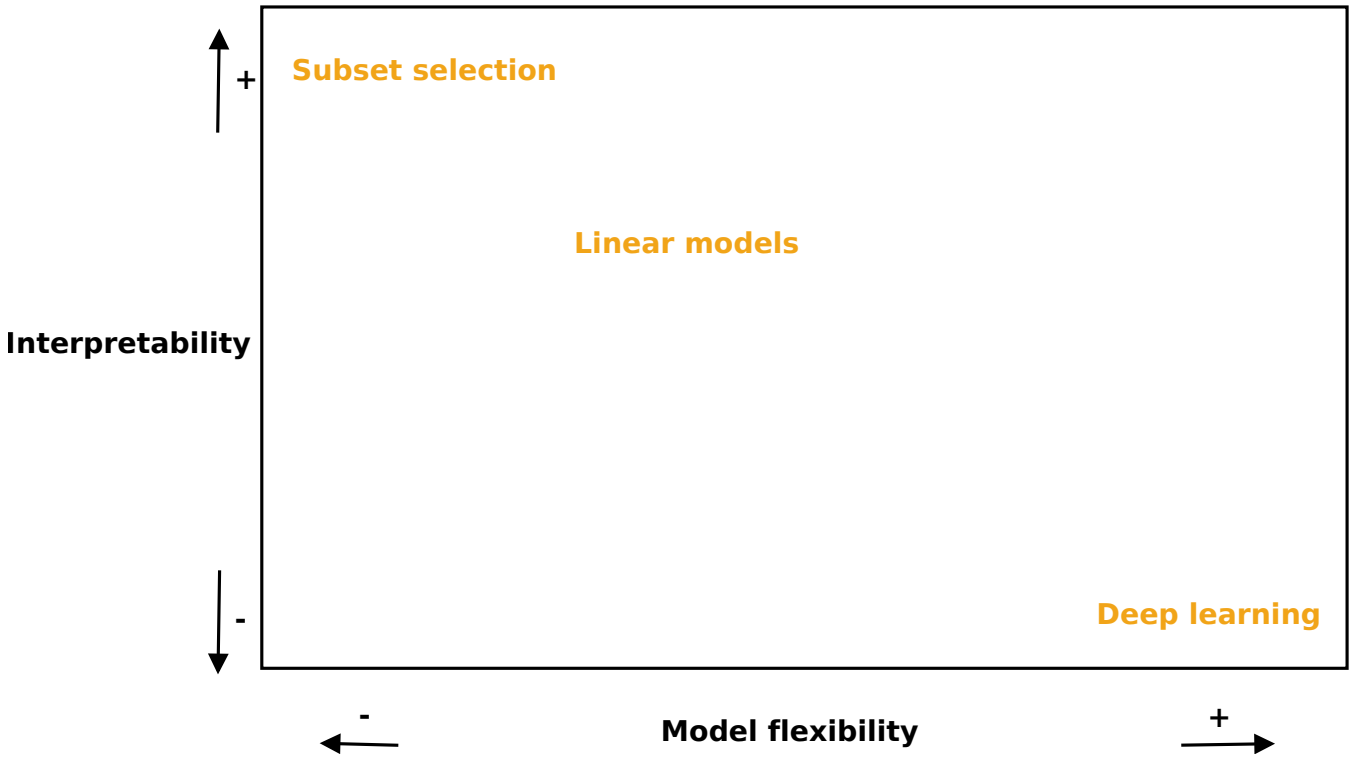
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Subset selection
Linear models
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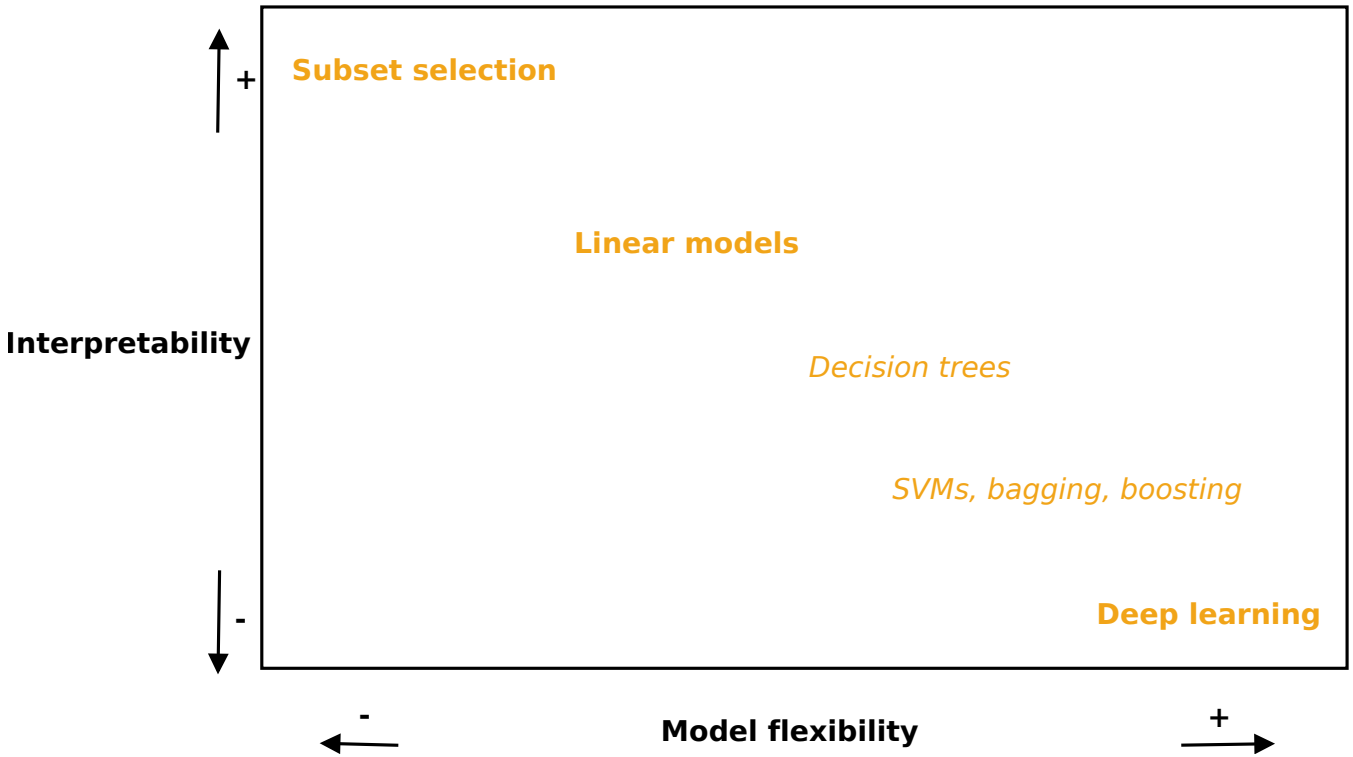

+
Subset selection
Linear models
Interpretability
Decision trees
SVMs, bagging, boosting
-
Deep learning
-
Model flexibility
+

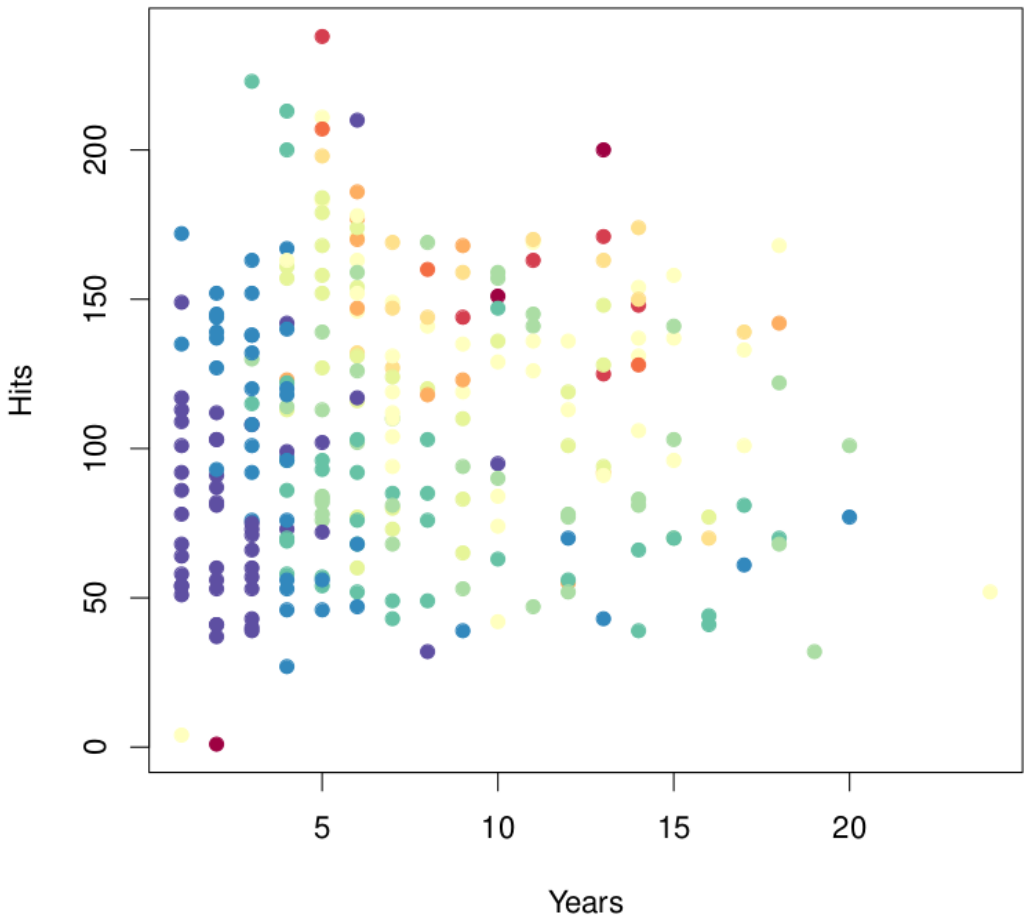
Classification and regression trees (CARTs)

- They work based on *segmentation* of the input space $\mathcal{X}$ into a number of simple regions
- It is done through a set of splitting rules
- The *decision tree* summarizes all possible sequences of sppliting rules for a given segmentation

More on decision trees

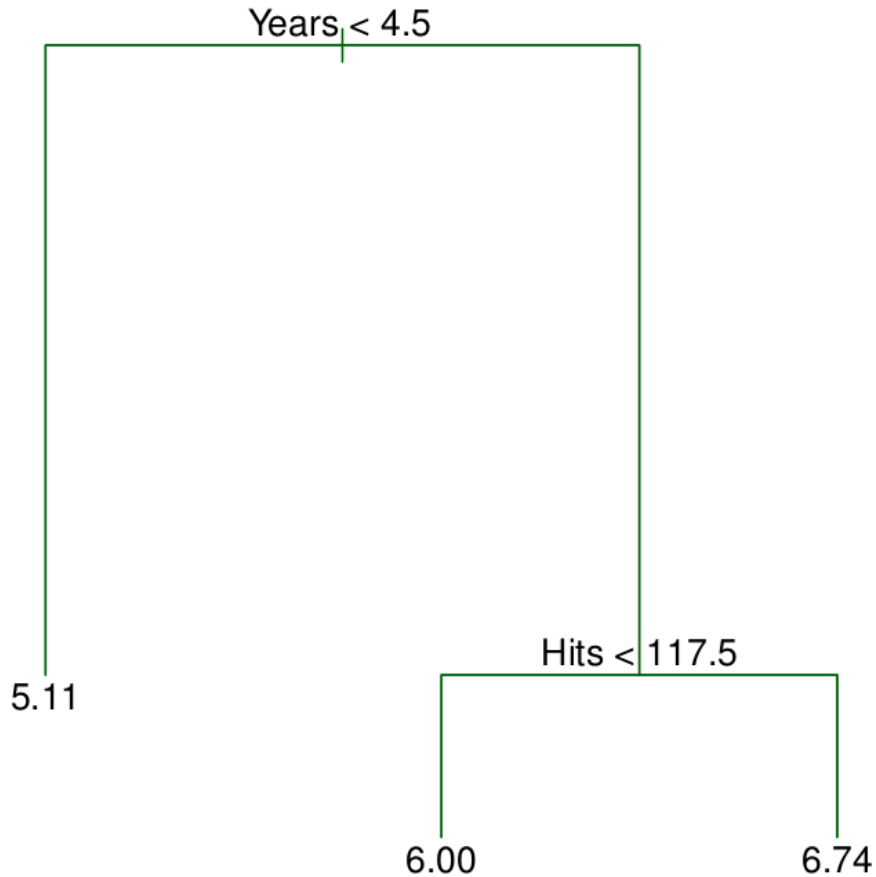
- They are simple and of easy interpretation
- They can be used for both regression and classification problems
- Let's first see them working in regression

Example: Baseball dataset



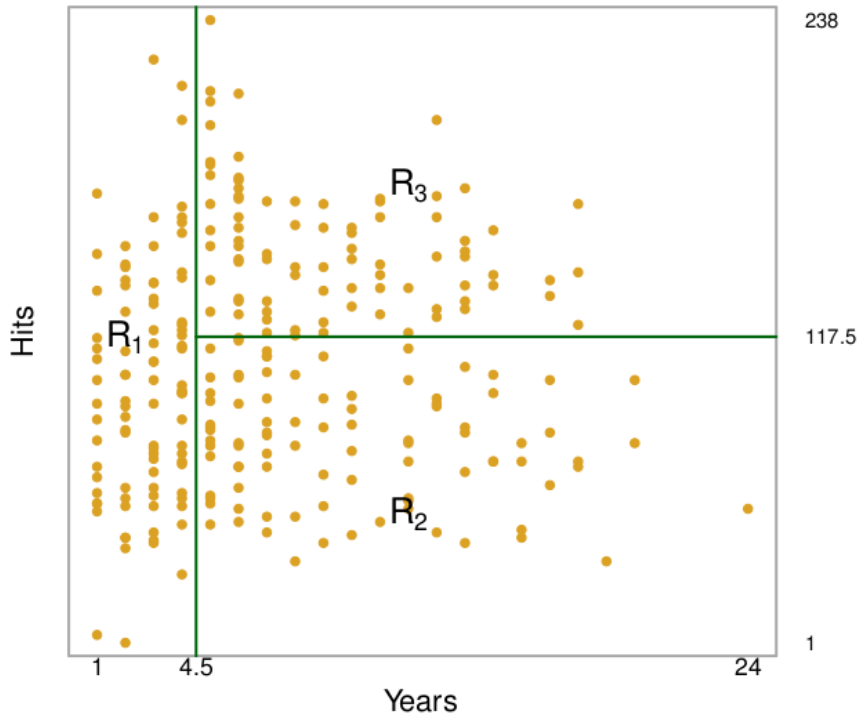
Salary goes from low (blue, green) to high (yellow, red)

Decision tree for `Baseball` dataset



- Leaf values are the mean of the response for observations that fall there

Yielded segmentation for Baseball dataset

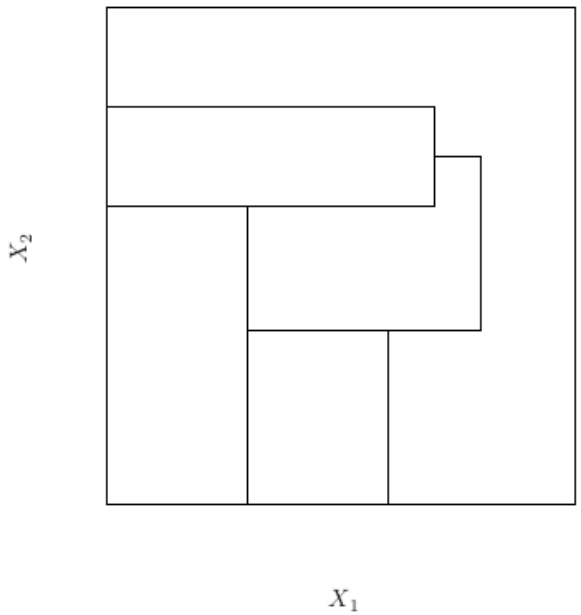


- The tree segments the players into three regions of $\mathcal{X}$:
  - $R_1 = \{x | \text{Years} < 4.5\}$
  - $R_2 = \{x | \text{Years} \geq 4.5 \text{ and Hits} < 117.5\}$
  - $R_3 = \{x | \text{Years} \geq 4.5 \text{ and Hits} \geq 117.5\}$

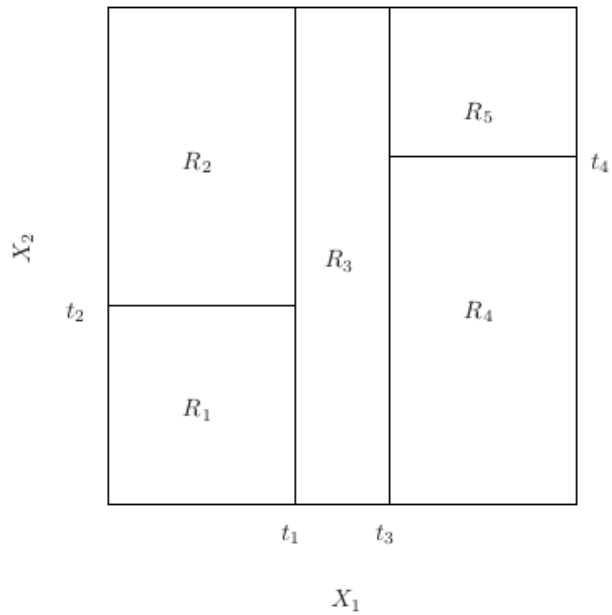
Basics of decision trees

- We split $\mathcal{X}$ into J non-overlapping regions $R_1, R_2, \ldots, R_J$
- For every observation (data point) that falls in R_j we make the same prediction (the mean of response values for the training observations in R_j)
- We generally segment $\mathcal{X}$ into a set of R_j high-dimensional rectangles (boxes), which make model intepretation easier

Data segmentation into boxes



Non-box segmentation



Box segmentation

Learning of CARTs

The goal is to find boxes $R_1, R_2, \ldots, R_J$ such that:

$$\min_{R_1,\ldots,R_J} \sum_{j=1}^{J} \sum_{i \in R_j} (y_i - \hat{y}_{R_j})^2, \tag{8.1}$$

where $\hat{y}_{R_j}$ is the mean response for all data points in the jth box R_j

- Procedure for learning the optimum set of boxes?

Learning $R_1, R_2, \ldots, R_J$

- Consider all possible partitions is not computationally feasible
- We need to use some suboptimal procedure
- A common one is a top-down, greedy approach

Top-down, greedy approach for learning the regression tree

1. Select variable x_j and cutpoint s_1 such that splitting $\mathcal{X}$ into $\{x|x_j < s_1\}$ and $\{x|x_j \geq s_1\}$ leads to the largest possible reduction in Eq. 8.1

2. In second iteration, we look for the x_k and cutpoint s_2 that splits **one of the previous two regions** in a way that leads to the greatest possible reduction in Eq. 8.1

3. This process continues until a stop criterion is achieved (e.g., no R_j has more than five data points)

Testing the trained tree

- The top-down, greedy approach has a shortcoming
  - It might overfit the data (segmentation memorizes the points)
- **One possible remedy:** work with few boxes $R_1, \ldots, R_j$
- This remedy, however, can lead to bad results if we are unluck to have a "bad" split in earlier iterations and latter ones to have "good" ones
- **Better alternative:** regularization through pruning

Regularization through pruning

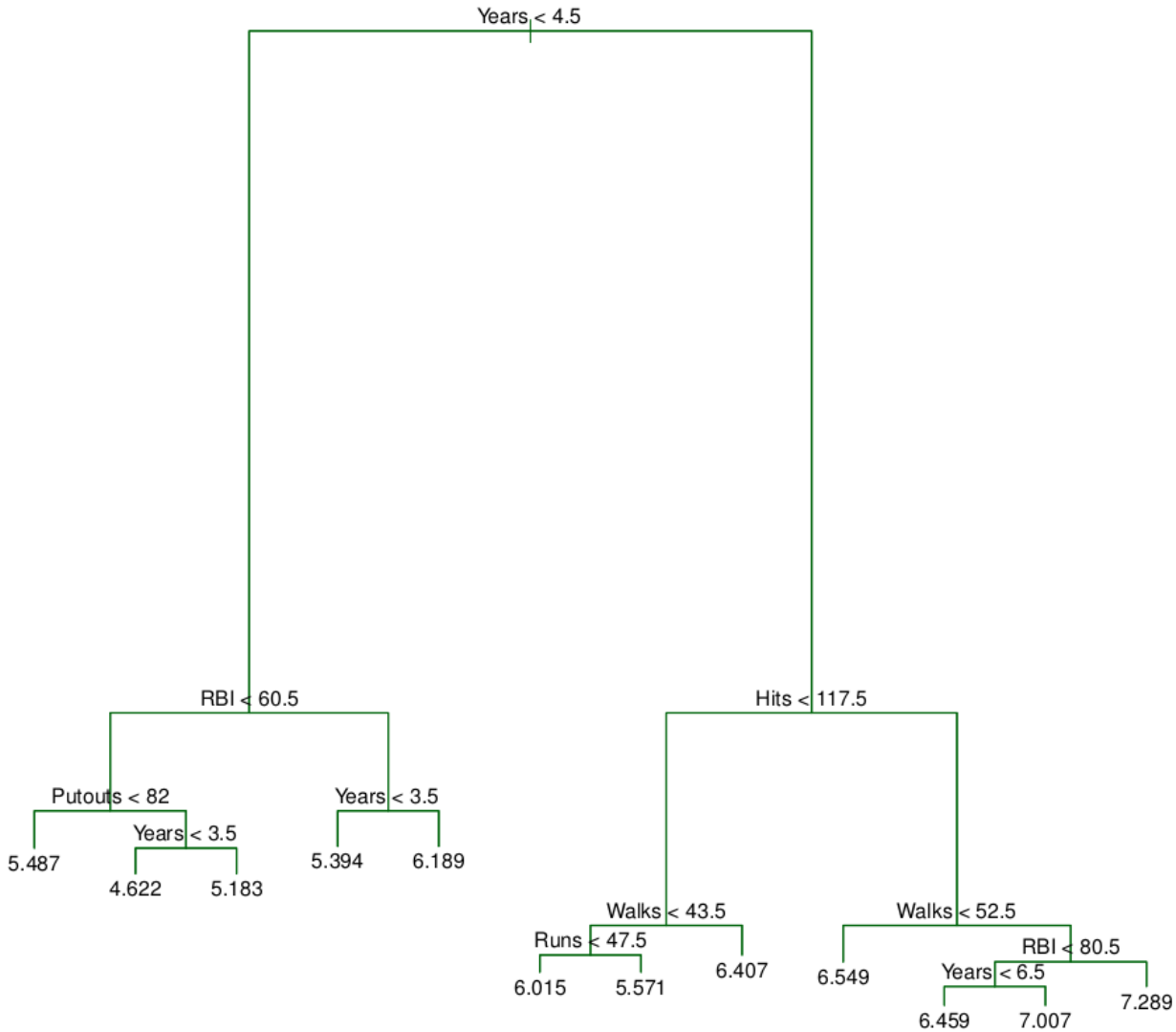
- Let $|T|$ be the number of leaves of tree T
- We compute a tree T_0, where $|T_0|$ is very large, and then prune it back to get a subtree $T \subset T_0$
- Given T_0, the size of T is chosen by minimizing the expression:

$$\min_m \sum_{m=1}^{|T|} \sum_{i:x_i \in R_m} (y_i - \hat{y}_{R_m})^2 + \alpha |T|, \tag{8.4}$$

where α is a hyperparameter

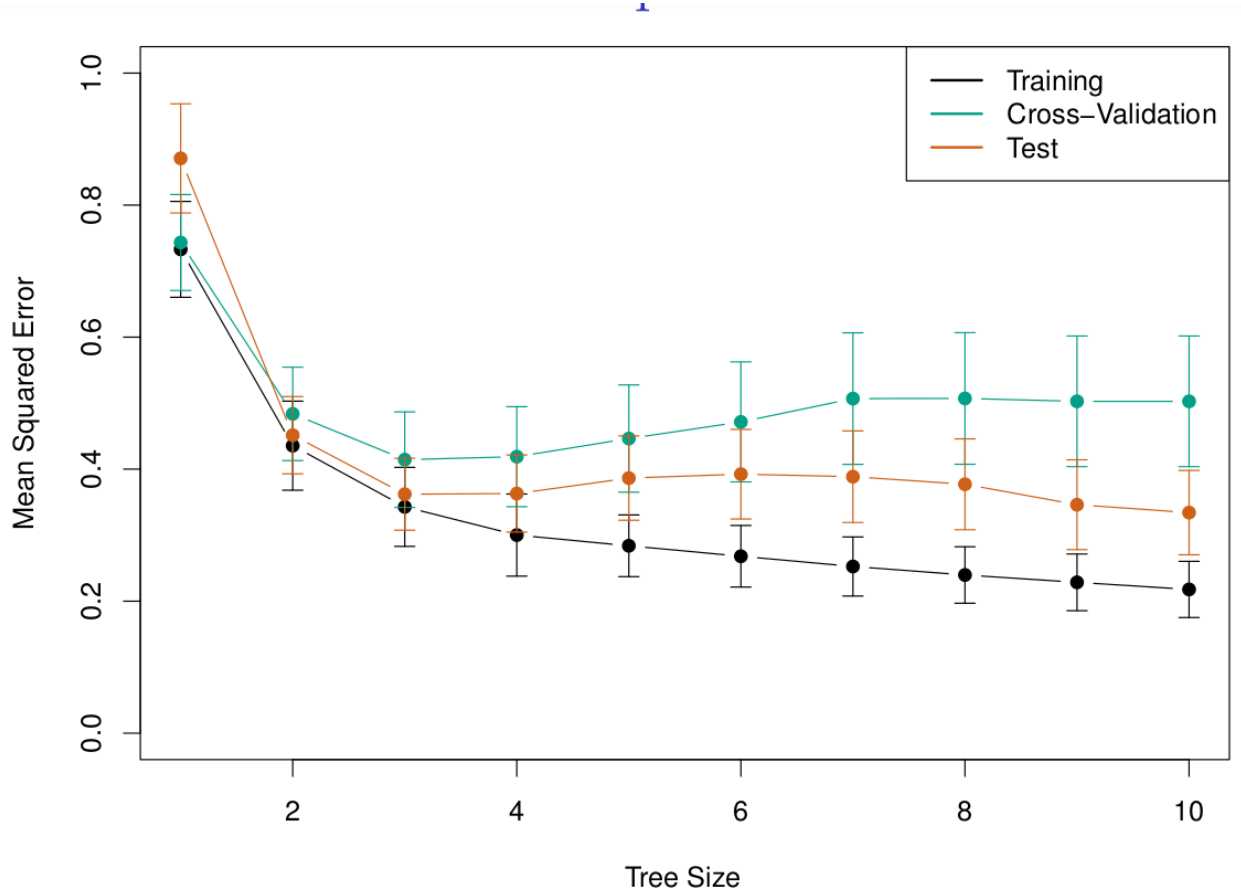
- We can use K-fold CV to choose a suitable value for α

Example: Baseball dataset using more attributes

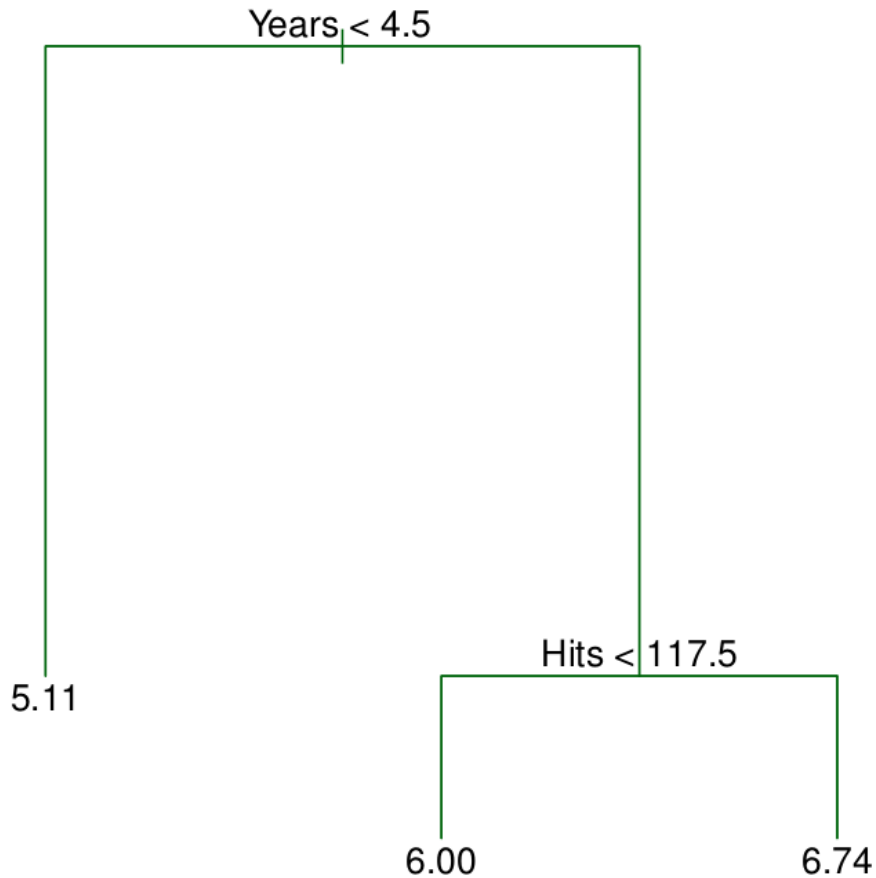


$$x = (\texttt{Hits RBI Runs Putouts Walks Years})^T$$

6-fold CV on Baseball



Chosen pruned tree T ($|T| = 3$)



Classification trees

- Used to predict a qualitative response instead a quantitative one

- We predict that a data point belongs to the most commonly occuring class in the segment R_j to which it belongs

- Instead of Eq. 8.1, we use a classification error rate, defined as the fraction of the training observations that don't belong to the most common class:

$$E = 1 - \max_k(\hat{p}_{mk}), \tag{8.5}$$

where $\hat{p}_{mk}$ is the proportion of data points in mth region that are from the kth class

Two other error measures

- Classification error rate is not sensitive enough for the tree-growing step, so in practive we use other measures

- The *Gini index* is given by:

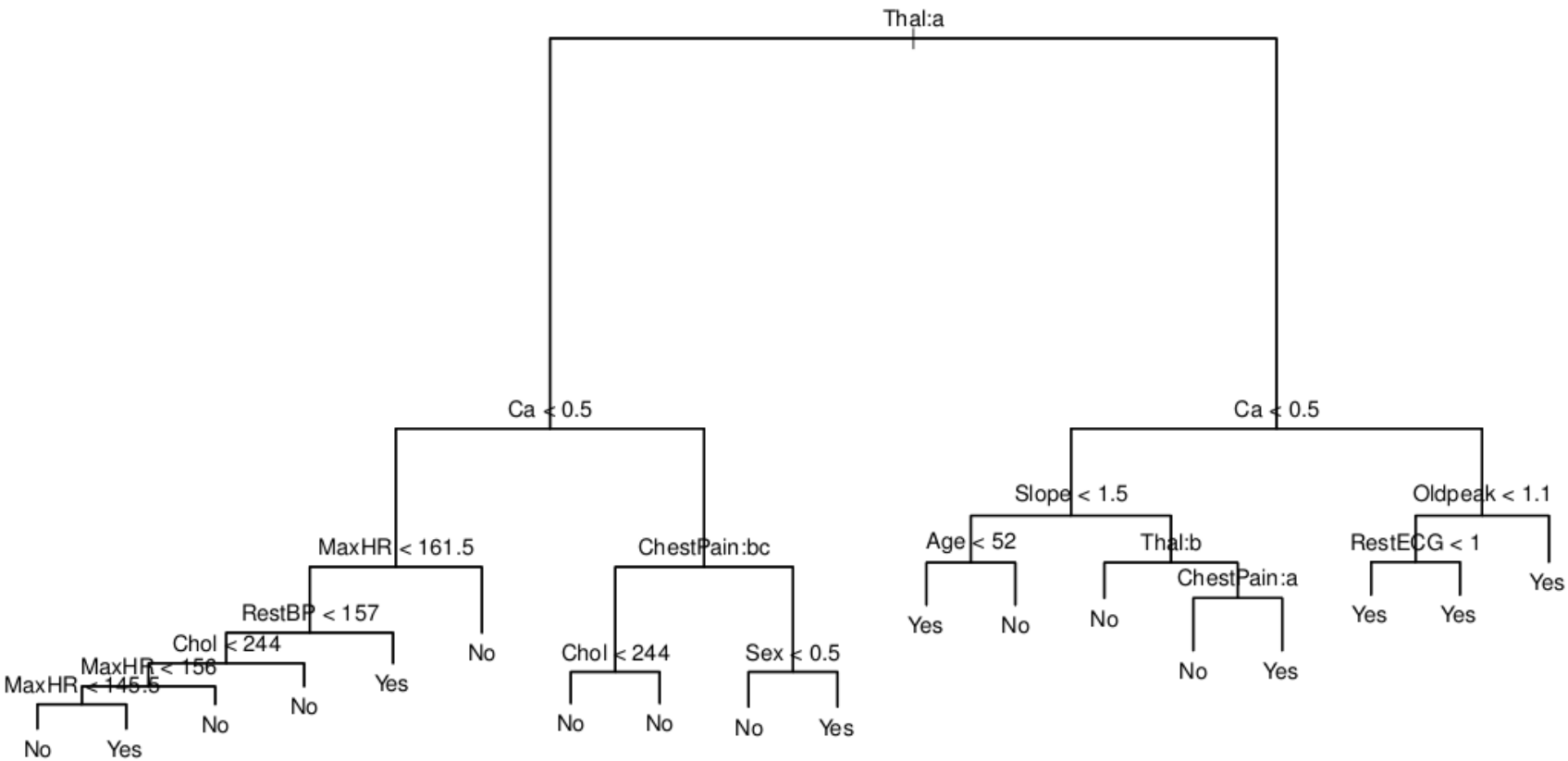
$$G = \sum_{k=1}^{K} \hat{p}_{mk}(1 - \hat{p}_{mk}), \tag{8.6}$$

  which is a measure of total variance across the K classes.

- Other possibility is the usage of *cross entropy*:

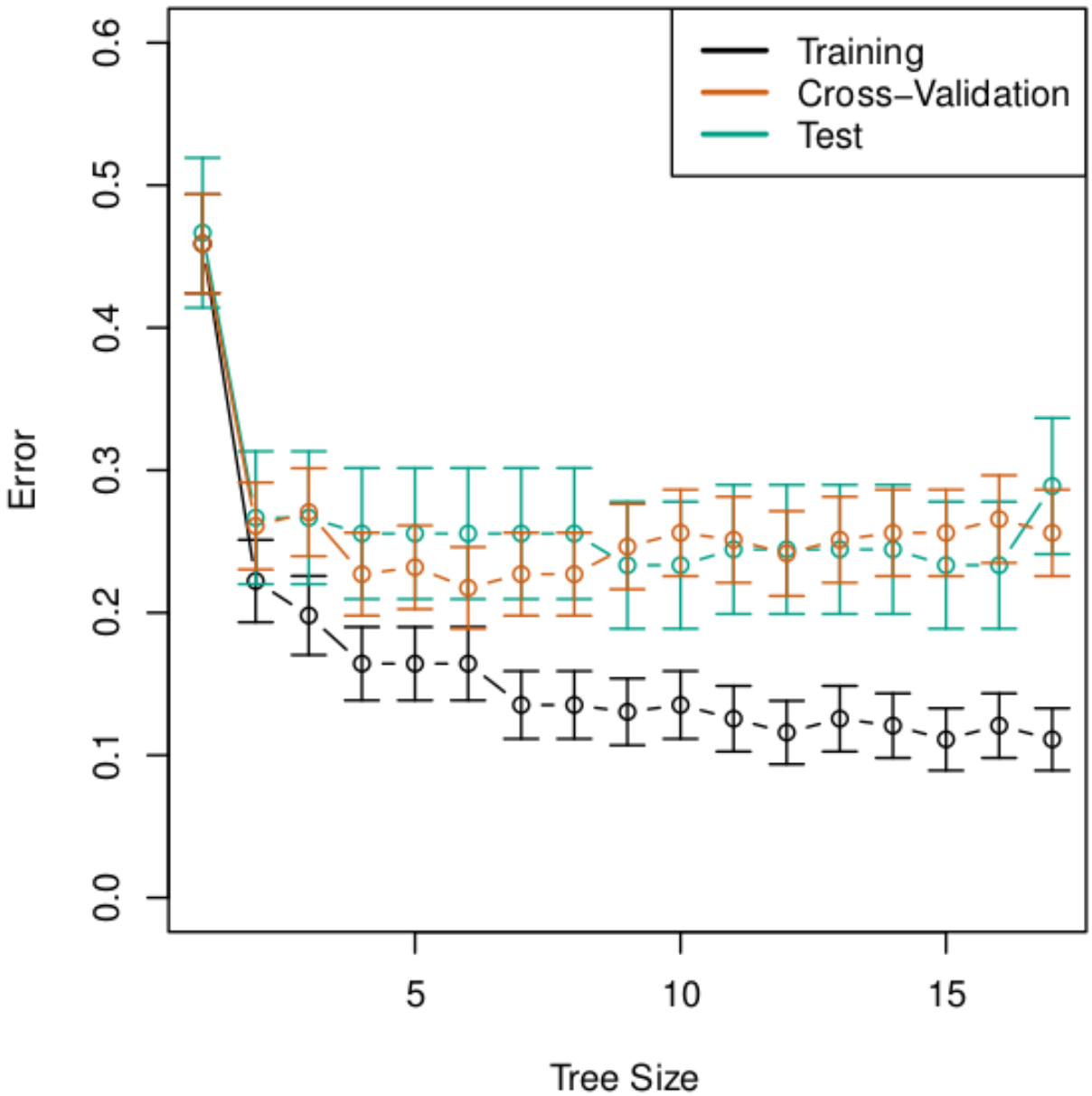
$$D = -\sum_{k=1}^{K} \hat{p}_{mk} \log \hat{p}_{mk} \tag{8.7}$$

Example: `Heart` classification dataset

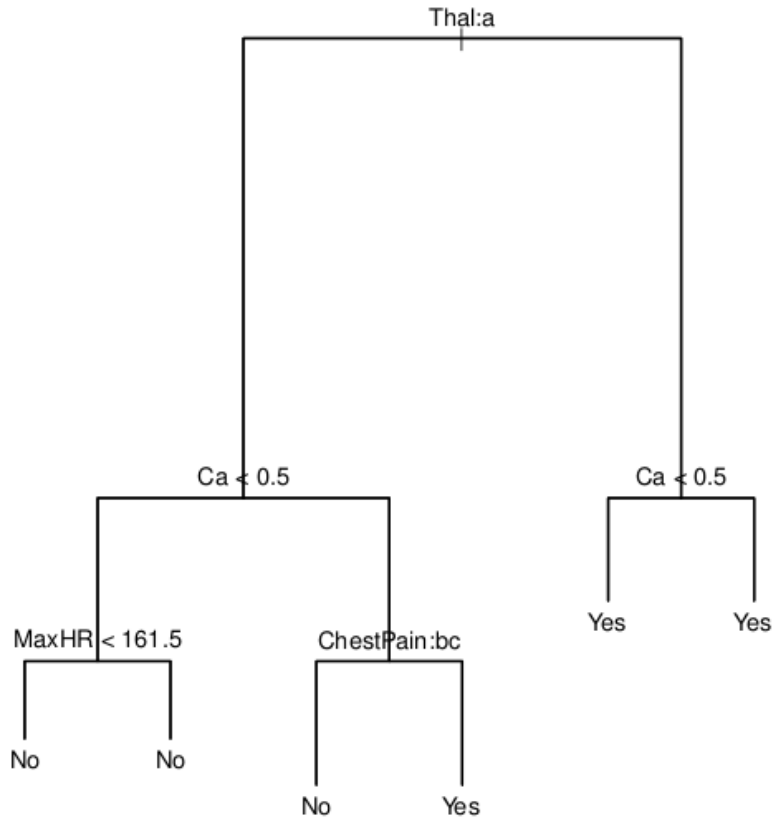


The full T_0 tree

Example: Heart classification dataset

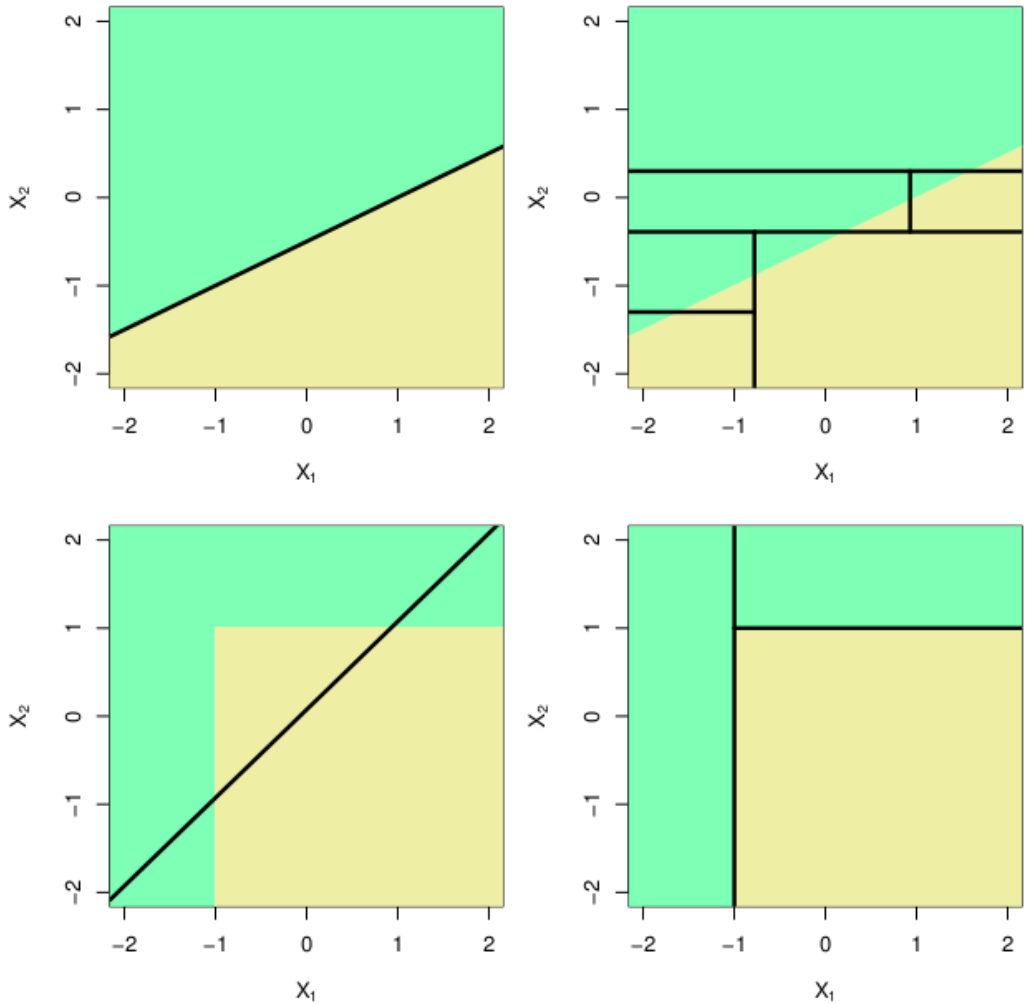


Example: `Heart` classification dataset



The chosen T through CV

Linear regression vs. decision trees



- True processes boundaries given in colors
- Left column: linear model; Right column: decision trees

Limitations of decision trees

- We saw that trees have very good interpretability
- They can be displayed graphically and are good to deal with qualitative attributes
- However, they often don't have a good accuracy
- Workarounds to improve predictive performance?

Ensemble methods

- An ensemble model use **multiple learning models** instead of a single one

- It often yields better predictive performance than using any of its composing models alone

- Some types of ensemble methods:

  1. Bagging
  2. Random forest
  3. Boosting

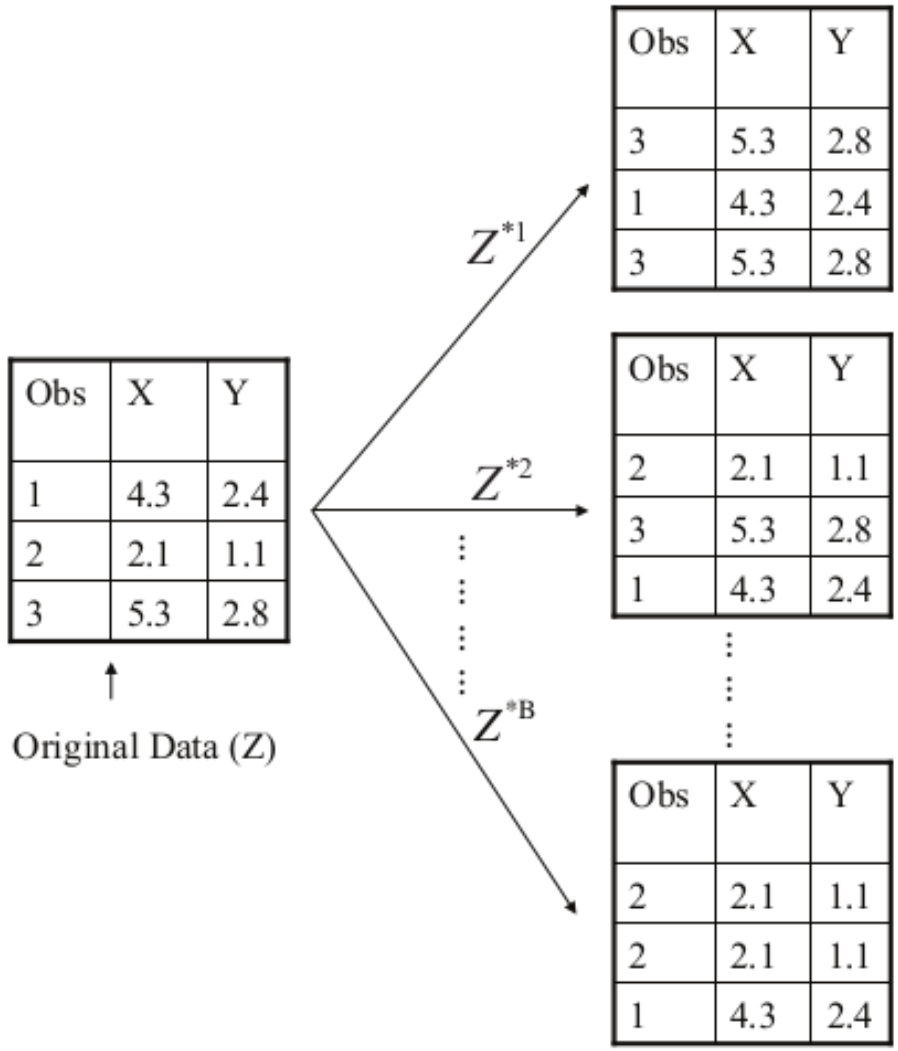
1. Baggings

- Also known as "bootstrap aggregations", they are a method for reducing the variance of a learning model
- They are a general-purpose method, however especially useful in the context of decision trees
- They are based in the fact that averaging results from different datasets reduces model variance

Obtaining more datasets

- How to obtain more datasets?
- We can use an approach called **bootstrap**
- In bootstrap, we take repeated samples *with replacement* from our training dataset to generate new datasets

Bootstrap example for $N = 3$ datapoints (observations)



Designing a bagging model

1. We produce B bootstrapped datasets

2. The bth bootstrapped training set is used train the classifier/regression $\hat{f}^{*b}(x)$

3. Final result is obtained by taking the most often classification among $\hat{f}^{*1}(x), \ldots, \hat{f}^{*B}(x)$ or, for regression, by averaging all the predictions:

$$\hat{f}_{bag}(x) = \frac{1}{B} \sum_{b=1}^{B} \hat{f}^{*b}(x)$$

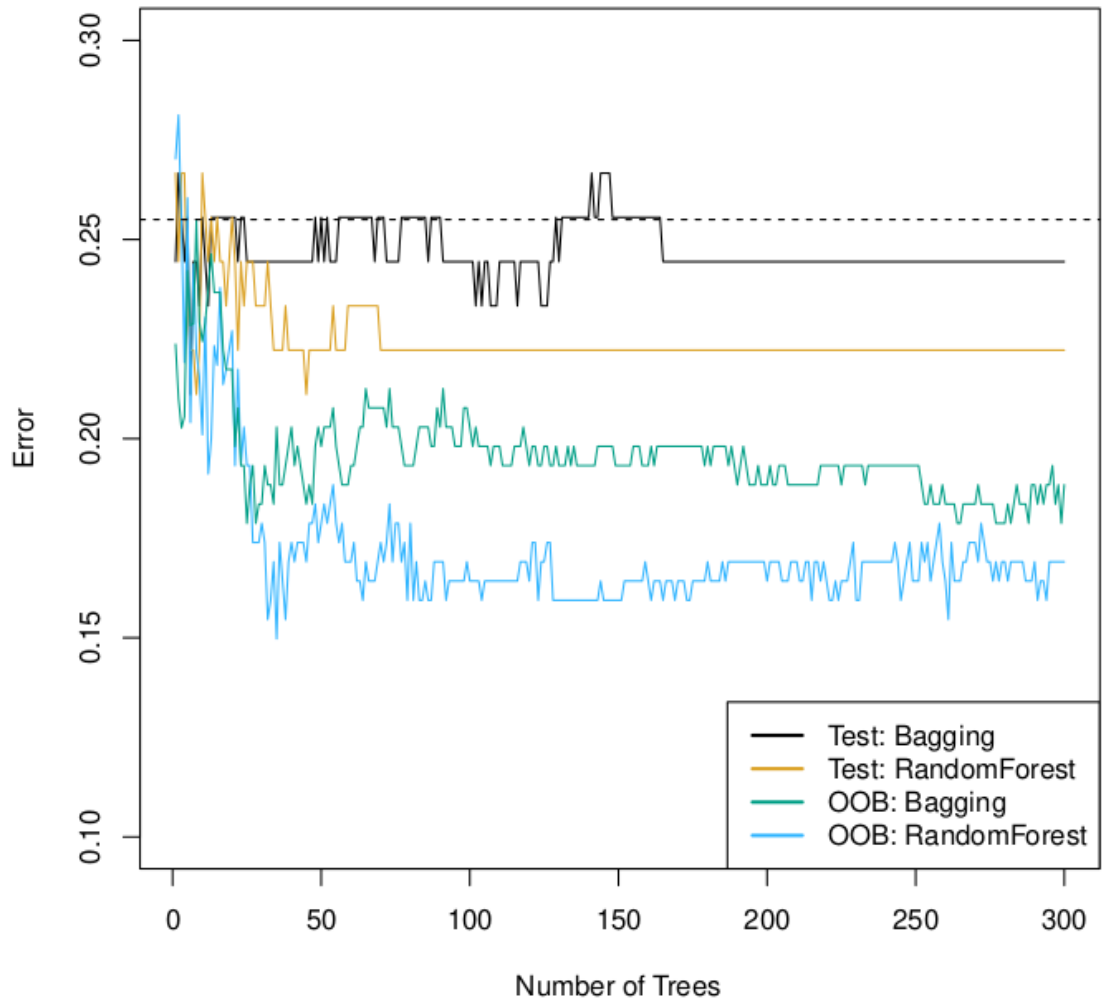
Out-of-bag error estimation

- If we generate B bootstrapped datasets Z^{*b} using a dataset Z of size n, one can show that, on average, each Z^{*b} makes use of $\frac{2}{3}n$ observations
- That means that, on average, each bagged tree doesn't use $\frac{1}{3}n$ observations for training
- For each x in Z, we can classify it using bagged trees that don't use x for training and average the results
- For a large value of B, this procedure is akin to leave-one-out CV

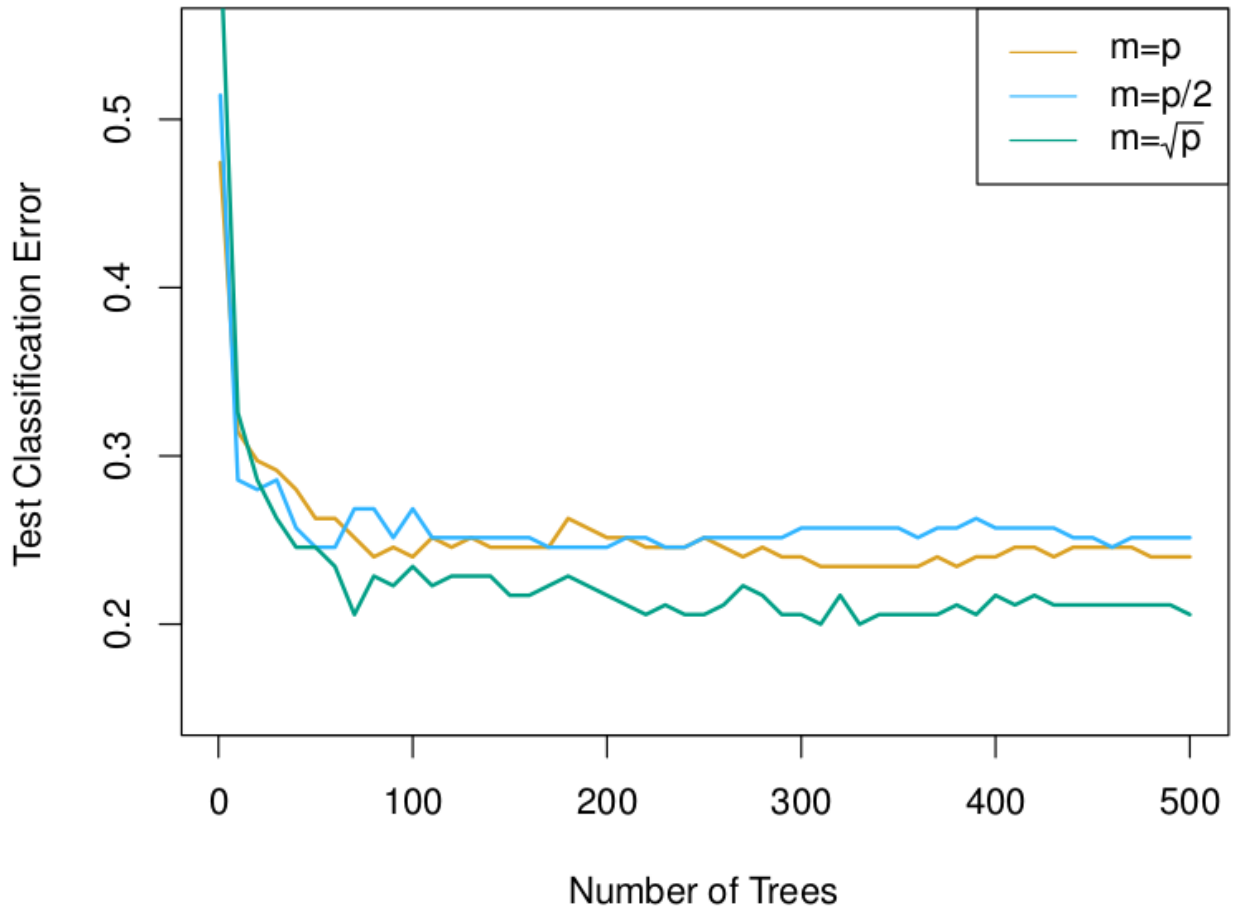
2. Random forests

- Improvement over bagged trees, based on a change in the tree-building process that decorrelates the trees
- This change reduces the variance when we average the trees
- The change is: at each iteration of the tree building process, a random subset of $x_1, \ldots, x_p$ is chosen, and the best split is defined only using variables of that subset
- Typical subset size is $\sqrt{p}$

Example 1: Heart dataset



Example 2: Gene expression dataset



3. Boosting

- Also a general general-purpose method that is very useful in the context of trees
- Instead of sampling the training dataset, Boosting uses a different strategy to build an ensemble
- In Boosting, the models are trained sequentially, using the information from previous models

3. Boosting

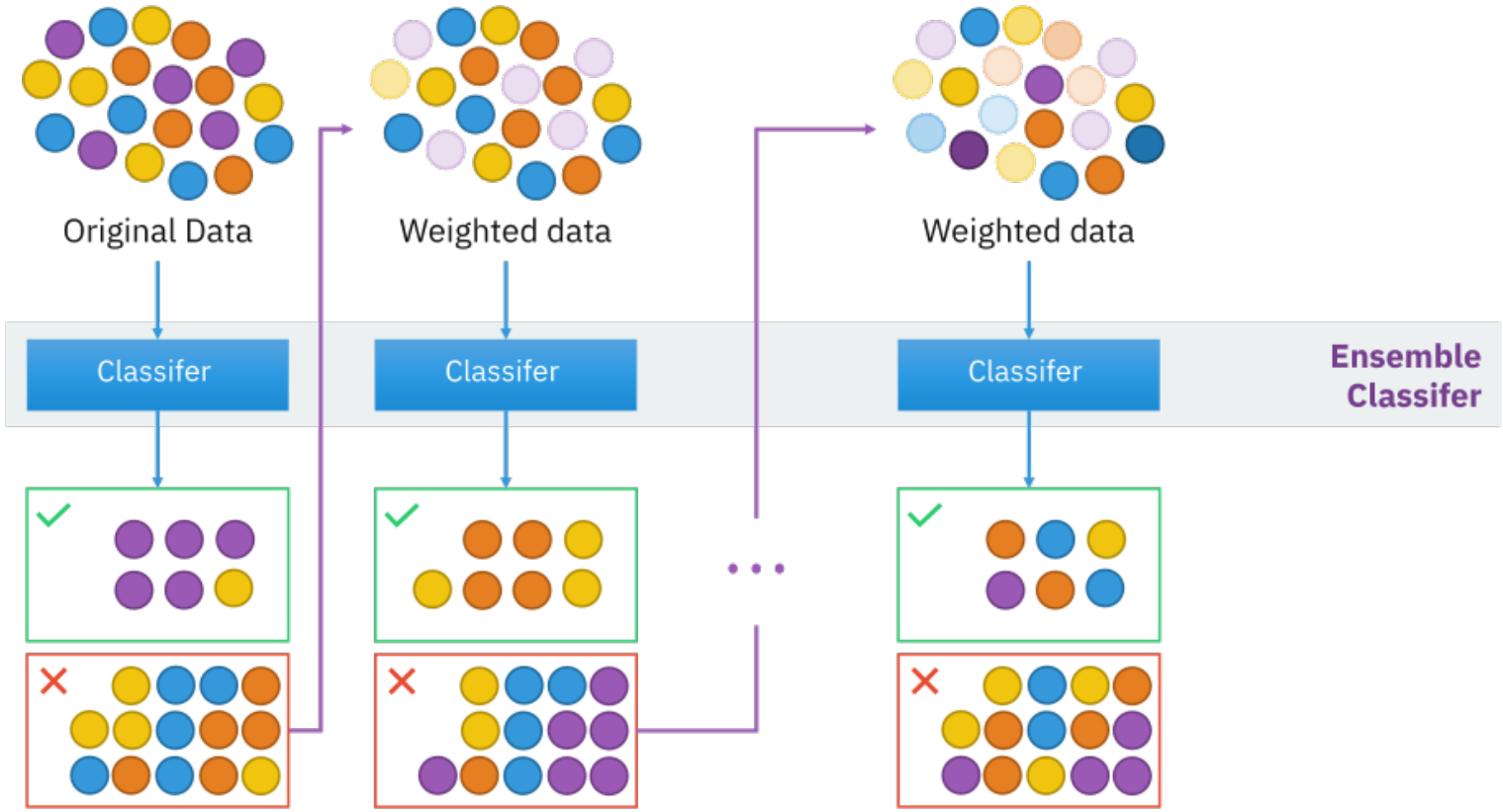


Figure: Boosting Ensemble scheme by Sirakorn, CC BY-SA 4.0, wikimedia id = 85888769

3. Boosting

- Boosting approach learns slowly. The trees are usually small and are determined by the parameter d.
- Given the current model, we fit a decision tree to the residuals from the model. We then add this new decision tree into the fitted function to update the residuals
- The training process forces the models to improve in areas where the others have failed, focusing on difficult parts

3. Boosting

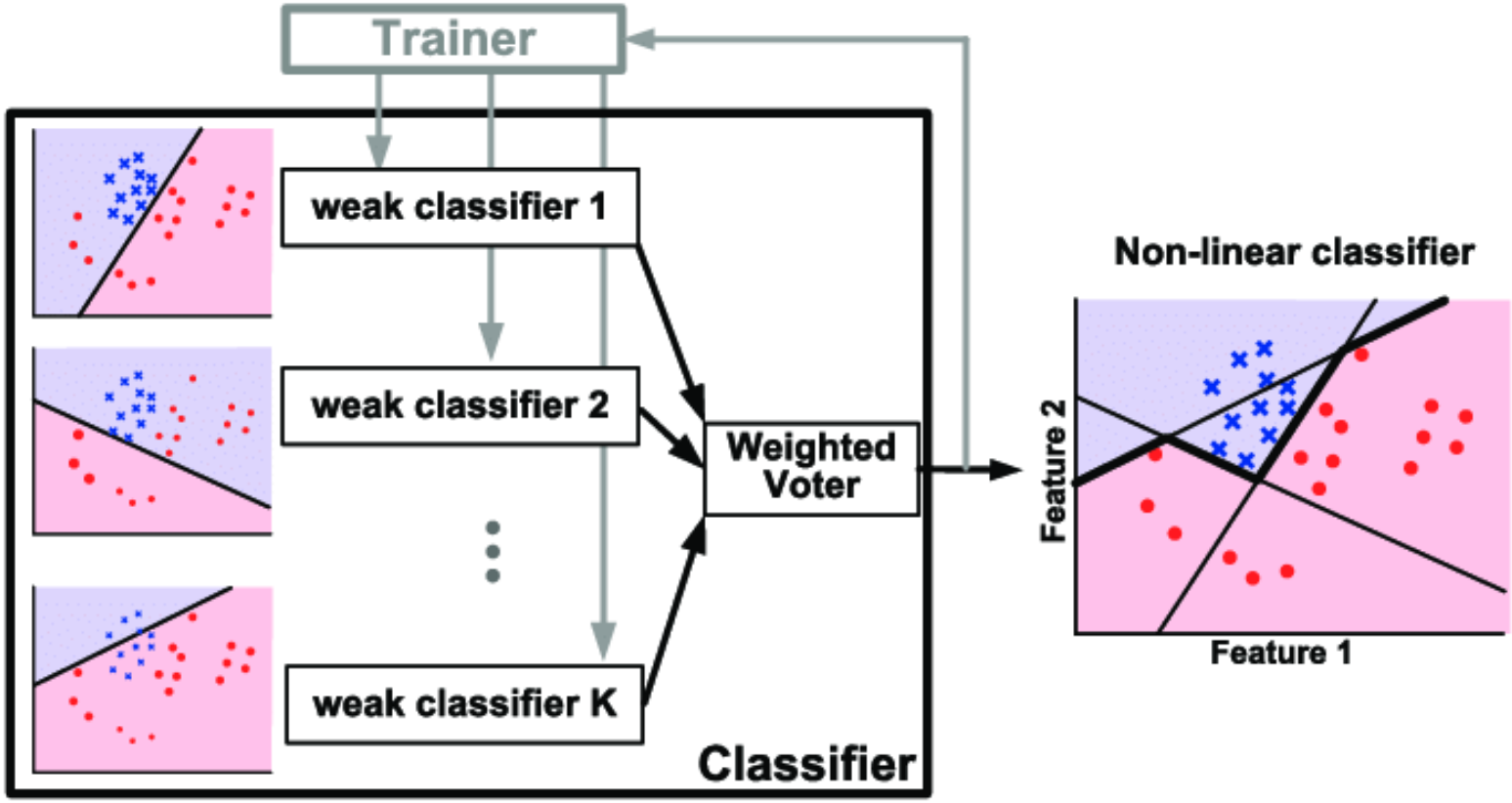


Figure: Result of fitting a boosting ensemble with weak classifiers in 2D feature space.

Realizing Low-Energy Classification Systems by Implementing Matrix Multiplication Directly Within an ADC - Scientific Figure on ResearchGate. Available from: https://www.researchgate.net/figure/Illustration-of-AdaBoost-algorithm-for-creating-a-strong-classifier-based-on-multiple_fig9_288699540 [accessed 29 Apr, 2024]

Algorithm 8.2: Boosting for Regression Trees

1. Set $\hat{f}(x) = 0$ and $r_i = y_i$ for all i in the training set.
2. For $b = 1, 2, ..., B$, repeat:
   a) Fit a tree $\hat{f}^b$ with d splits ($d + 1$ terminal nodes) to the training data (X,r).
   b) Update $\hat{f}$ by adding in a shrunken version of the new tree:
   $$\hat{f}(x) \leftarrow \hat{f}(x) + \lambda \hat{f}^b(x). \tag{8.10}$$
   c) Update the residuals,
   $$r_i \leftarrow r_i - \lambda \hat{f}^b(x). \tag{8.11}$$
3. Output the boosted model,
$$\hat{f}(x) = \sum_{b=1}^{B} \lambda \hat{f}^b(x). \tag{8.12}$$

Variable importance

- In those ensemble methods, we can measure the importance of each variable for classification/regression
- For regression, we store the total amount of reduction of the objective function during splits on a given variable
- For classification, we store the total amount that Gini index is decreased during splits over a given variable
- At either case, we average results over all B trees; a large value indicates an important variable

Variable importance example: `Heart` dataset